An Examination of the Relationship between Financial Characteristics and Portfolio Performance Using Models from Backward Elimination and PCA

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Inquiry of Interest

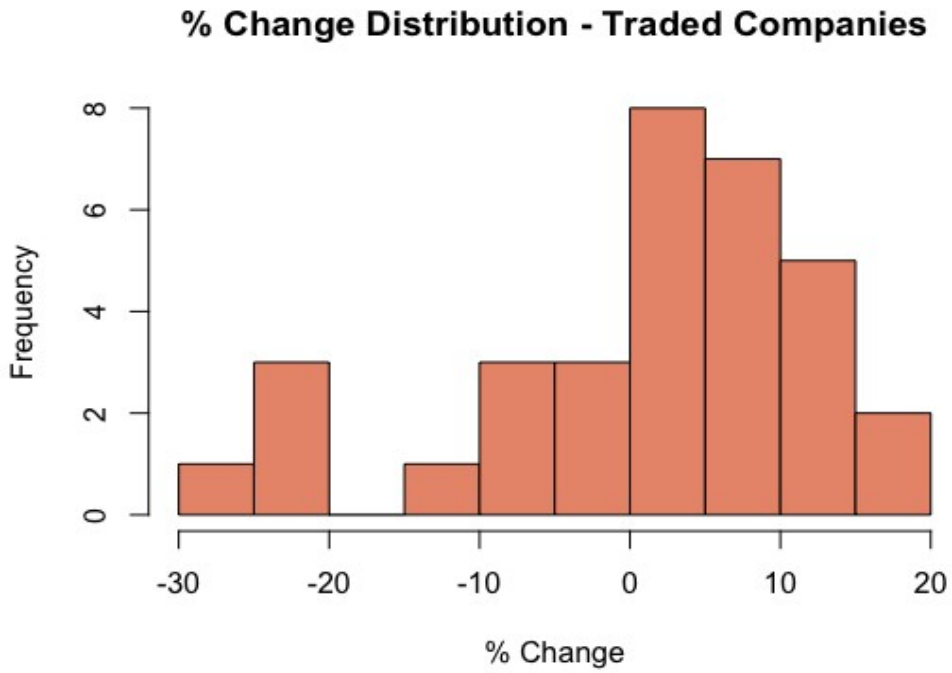
- Green Square Capital uses 13 factors to evaluate company performance
- Which factors are most important?
- We compare 2 models:
  - Backward elimination to reduce the number of predictor variables
  - Principal components to reduce the dimensions of predictor variables

The Data

- 33 traded companies across 8 sectors, 62 untraded
  - Consumer discretionary, consumer staples, energy, healthcare, information technology, materials, telecommunication services, utilities
- 13 variables per company
  - Price at start of quarter, price at end of quarter, cash/price, debt/equity, accounts receivable/revenue, total asset turnover, DPS quarter growth rate, DPS 5 year growth rate, dividend yield, EBITDA margin, unlevered FCF margin, price/earnings, price/book value, forward price/earnings, diluted EPS 5 year growth rate, EPS growth rate
- Response variable – Portfolio performance, we used percent change in price per share from 3/31 to 6/30 as a measure of portfolio performance

Backward Elimination

- Start with all predictors in the model
- Eliminate the predictor with highest p-value greater than α–criterion
- Stop when the adjusted R^2 starts to drop and the predictors meet the α–criterion
- Goal: explain the maximum amount of variance with the fewest number of predictors

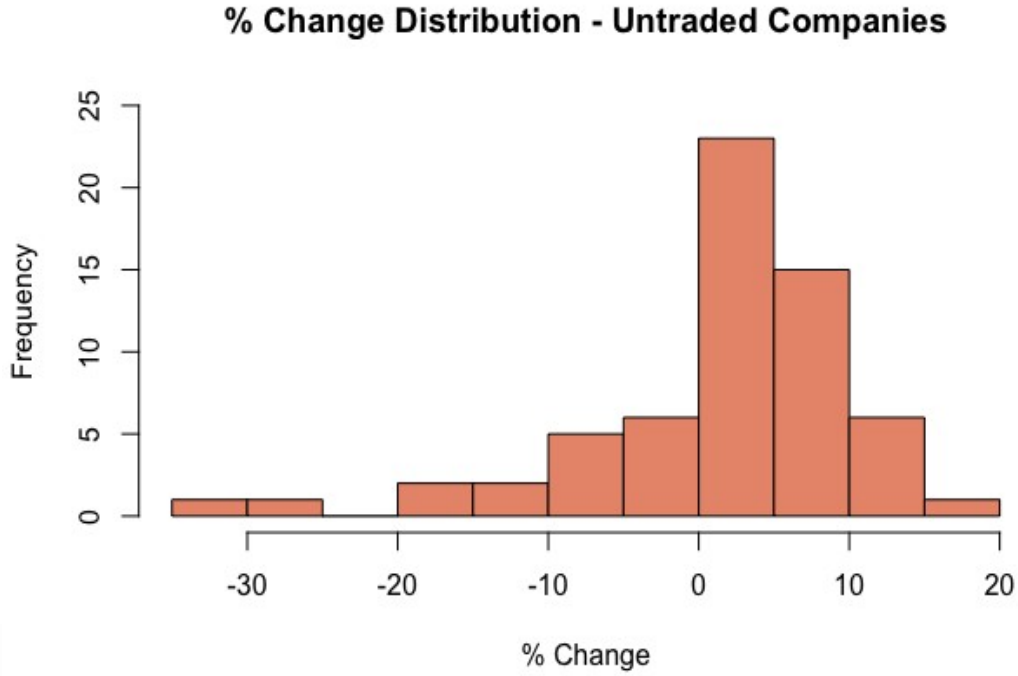


Mean and SD for Traded Companies:

Mean: .702
SD: 11.32

Mean and SD for Untraded Companies:

Mean: 1.32
SD: 8.99



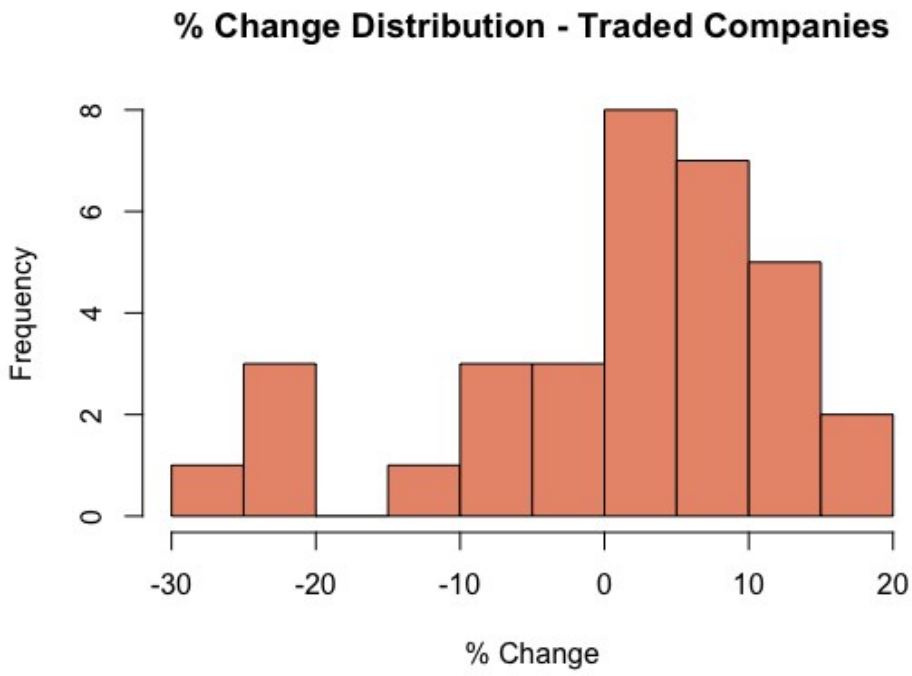
Backward Elimination Model

```
Call:
lm(formula = ds$Pct.Change ~ ds$DPS_5yrs + ds$DY + ds$TAT + ds$DEPS +
    ds$FCFM)

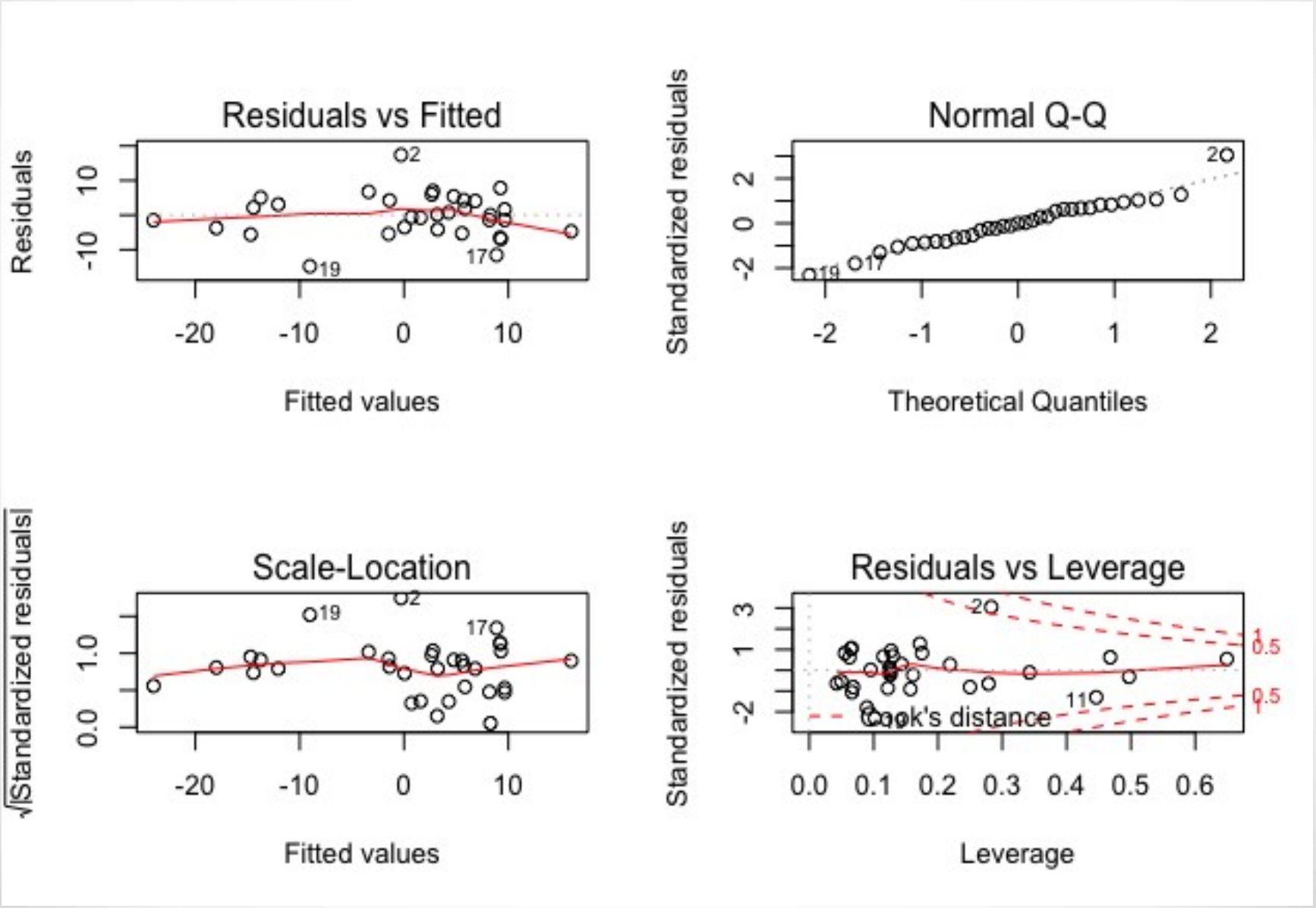
Residuals:
     Min       1Q   Median       3Q      Max
-14.7737  -4.1349   0.0178   4.2463  17.4000

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   23.214      5.330   4.355 0.000172 ***
ds$DPS_5yrs    2.837      1.440   1.970 0.059147 .
ds$DY         -5.855      1.048  -5.588 6.31e-06 ***
ds$TAT       -25.951     14.626  -1.774 0.087280 .
ds$DEPS       -3.350      2.540  -1.319 0.198323
ds$FCFM       11.282      6.365   1.772 0.087621 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.74 on 27 degrees of freedom
Multiple R-squared:  0.7011,    Adjusted R-squared:  0.6457
F-statistic: 12.66 on 5 and 27 DF,  p-value: 2.175e-06
```



Residual Plots



Backward Elimination Model w/out ADM

```
Call:
lm(formula = ds$Pct.Change ~ ds$PBV + ds$DPS_today + ds$DY +
    ds$TAT + ds$FCFM)

Residuals:
     Min       1Q   Median       3Q      Max
-11.4075  -1.8039  -0.2505   2.1661   9.4806

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   14.6796     4.5100   3.255 0.003143 **
ds$PBV         0.1516     0.1136   1.334 0.193684
ds$DPS_today   2.8734     0.8573   3.352 0.002467 **
ds$DY         -3.3580     0.7700  -4.361 0.000181 ***
ds$TAT       -61.9623     8.8657  -6.989 2.02e-07 ***
ds$FCFM       10.4869     4.6076   2.276 0.031321 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.924 on 26 degrees of freedom
Multiple R-squared:  0.8352,    Adjusted R-squared:  0.8035
F-statistic: 26.35 on 5 and 26 DF,  p-value: 2.091e-09
```



Note: when we remove company ADM, the Adjusted R^2 jumps from 64% to 80%. We also added predictors current DPS and Price/Book Value, and removed 5 year DPS and and Dividend Yield

Principal Component Analysis

- Principal Component Analysis (PCA) is a dimension reduction technique for regression
- Derives a low-dimensional set of features from a large set of variables known as principal components
- The goal : to explain the maximum amount of variance with the fewest number of principal components

Brief Intro on Computing Principal Components

- From our original data matrix, calculate the covariance matrix
- Find the associated eigenvalues and eigenvectors (orthonormal)
- Select largest eigenvalues and use the corresponding eigenvectors as our new basis
- These will be our Principal Components!
- Principal components are ranked by the proportion of variability in the data they explain
- In our analysis the first principal component explains 23% of the variability in our original dataset
- But how many more principal components should we consider…

Parallel Analysis

- R has a package called fa.parallel that we can use to determine how many components to keep
- fa.parallel simulates a dataset similar to the original and then creates a scree plot, which plots the variance explained by each principal component of both data sets in descending order
- This led us to consider only the first 4 principal components
- We want to create a model using the 4 principal components with a higher or comparable R^2

PCA Model

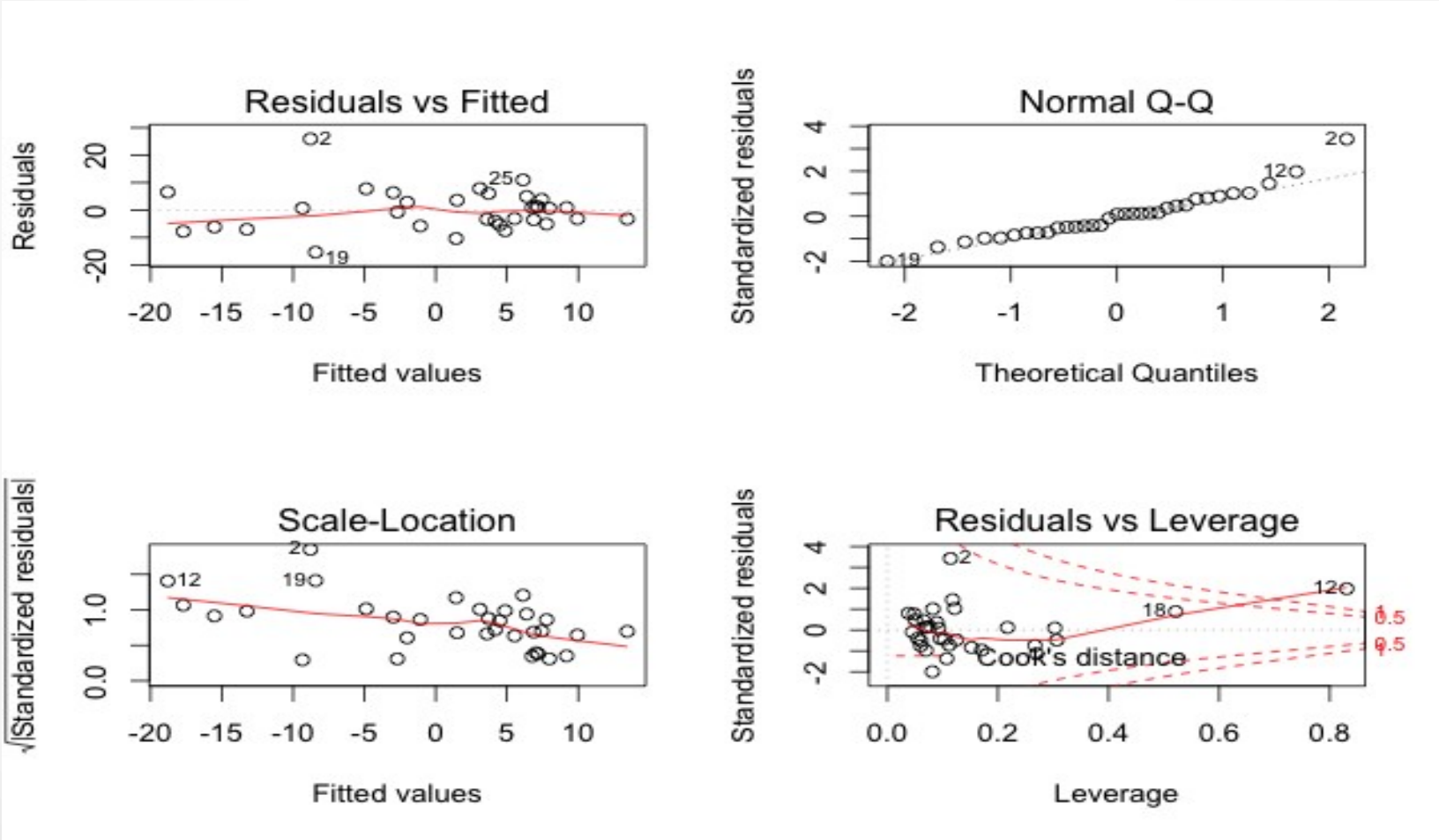
```
Call:
lm(formula = ds$Prop.Change ~ yv1 + yv2 + yv3 + yv4)

Residuals:
     Min       1Q   Median       3Q      Max
-15.3666  -5.1046   0.6393   3.8595  25.9066

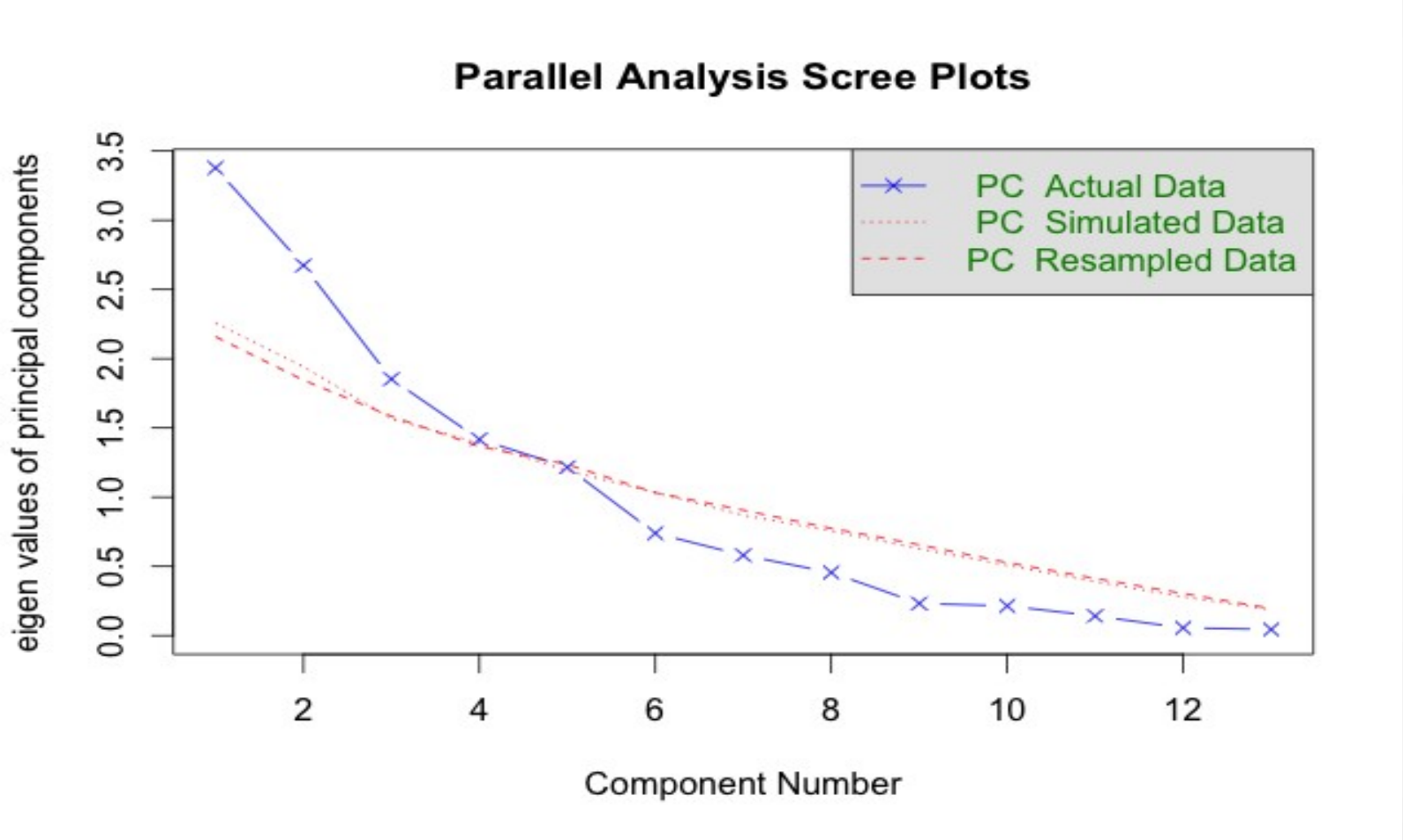
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.7026     1.3983   0.502 0.619285
yv1          -2.6341     0.7609  -3.462 0.001740 **
yv2           3.6051     0.8552   4.215 0.000235 ***
yv3          -0.6107     1.0273  -0.595 0.556935
yv4           2.7478     1.1747   2.339 0.026688 *
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

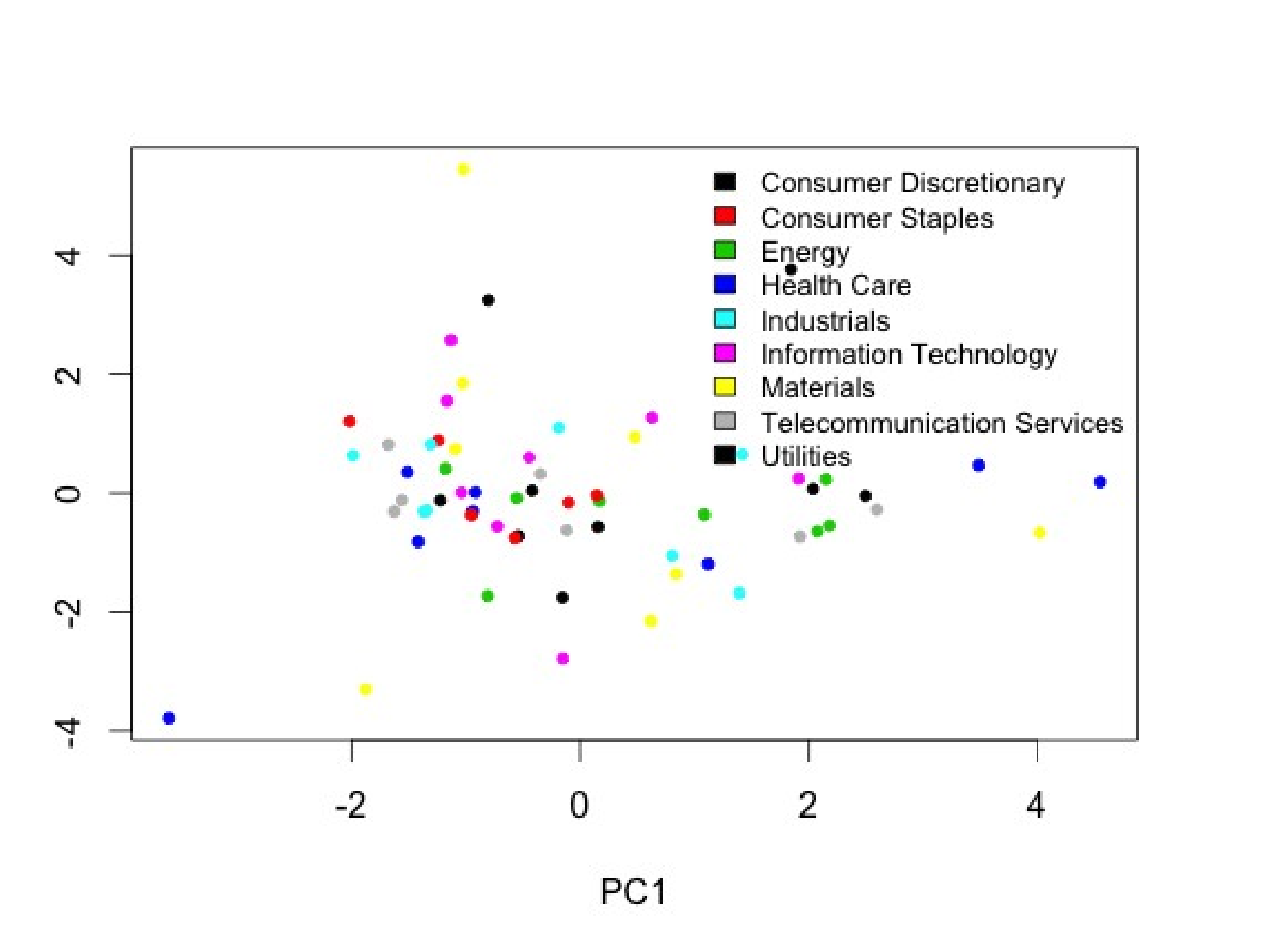
Residual standard error: 8.033 on 28 degrees of freedom
Multiple R-squared:  0.5596,    Adjusted R-squared:  0.4967
F-statistic: 8.895 on 4 and 28 DF,  p-value: 9.119e-05
```

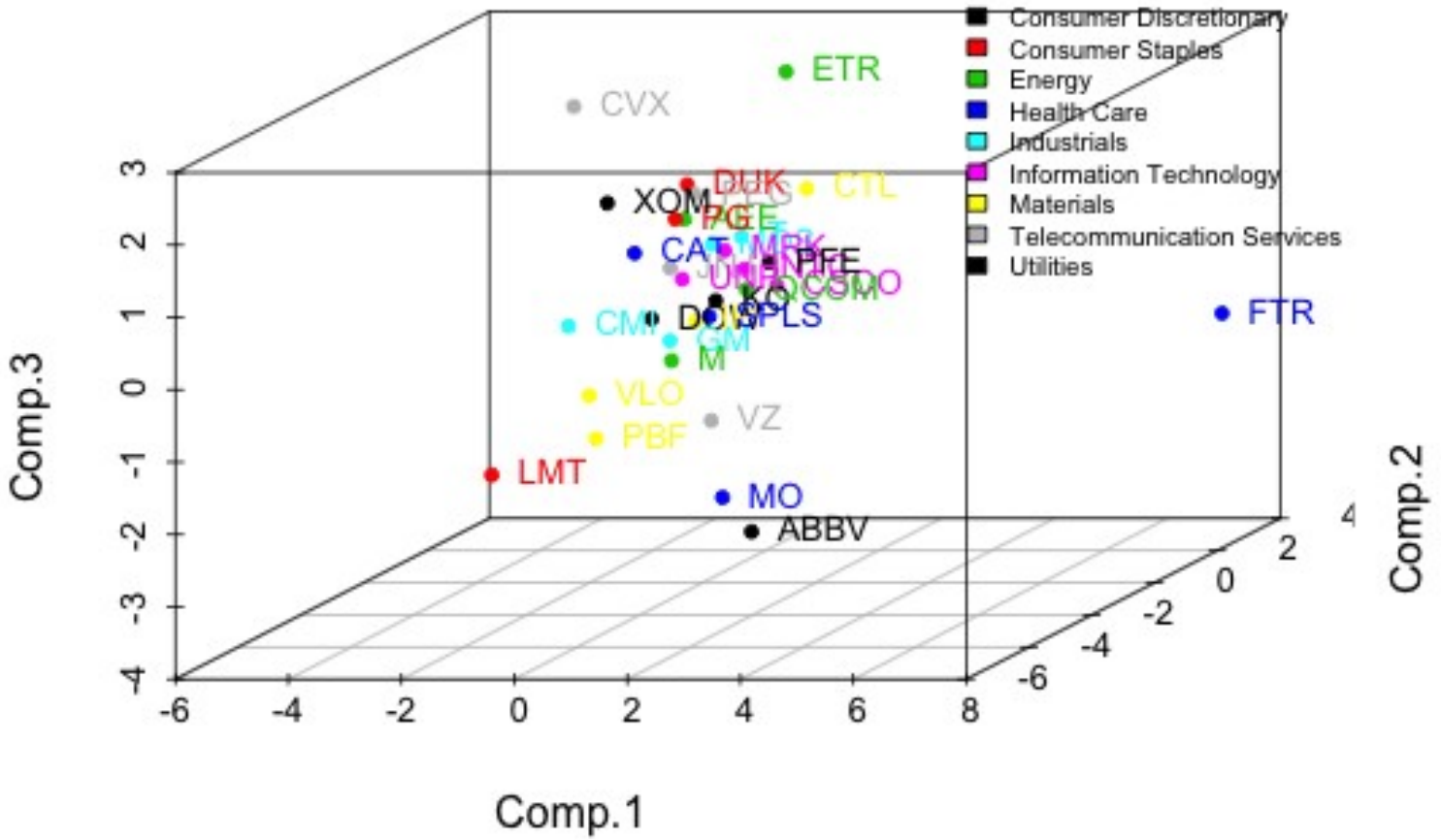
Residual Plots



Scree Plot – Traded Companies




Consumer Discretionary
Consumer Staples
Energy
Health Care
Industrials
Information Technology
Materials
Telecommunication Services
Utilities
-4
-2
0
2
4
-2
0
2
4
PC1


Consumer Discretionary
Consumer Staples
Energy
Health Care
Industrials
Information Technology
Materials
Telecommunication Services
Utilities
ETR
CVX
DUK
CTL
XOM
PCG
CAT
MRK
PFE
KO
SPLS
FTR
CMI
GM
M
VLO
VZ
PBF
LMT
MO
ABBV
Comp.1
Comp.2
Comp.3

PCA Model w/out ADM

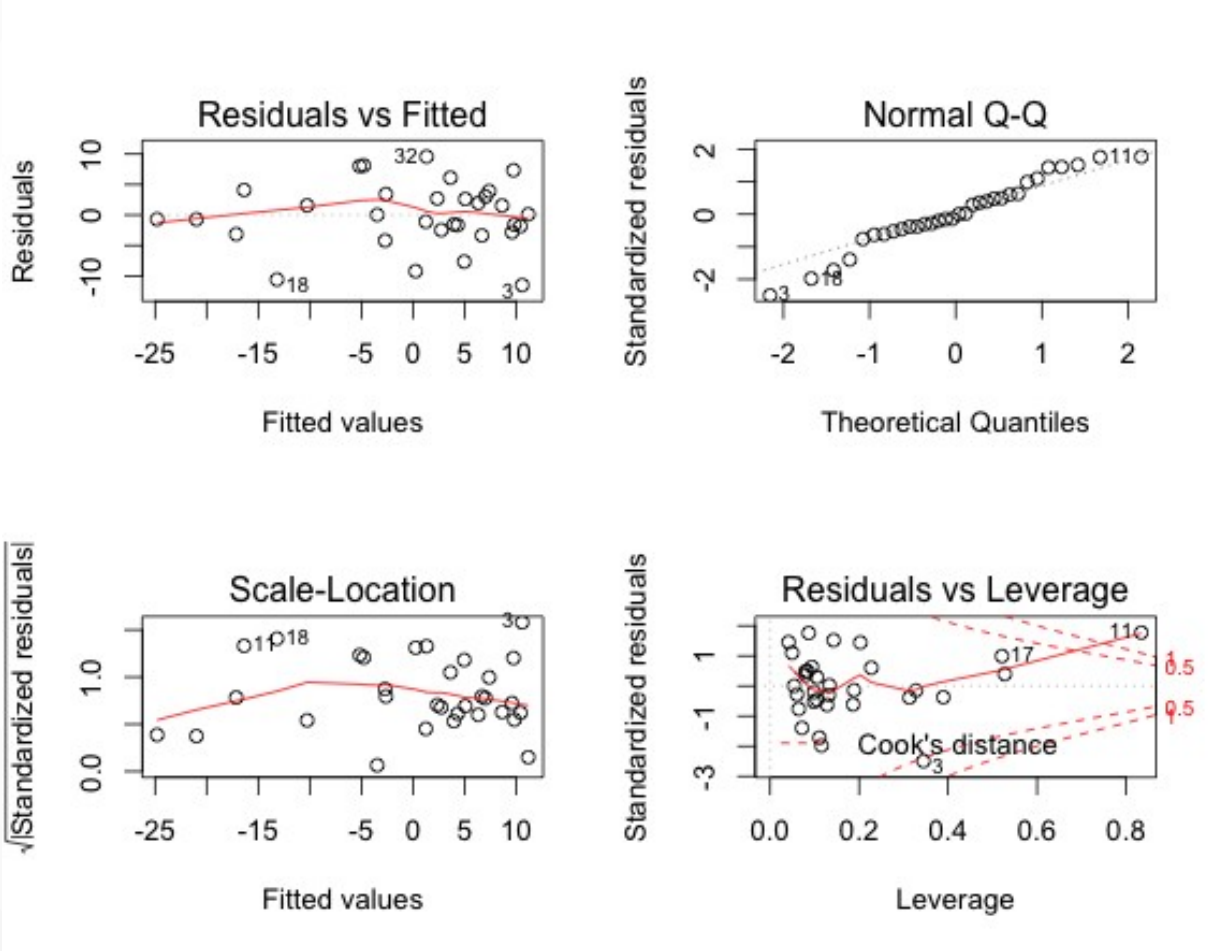
```
Call:
lm(formula = ds$Prop.Change ~ yv1 + yv2 + yv3 + yv4 + yv5)

Residuals:
     Min       1Q   Median       3Q      Max
-11.5143  -2.5672  -0.3381   3.1240   9.5818

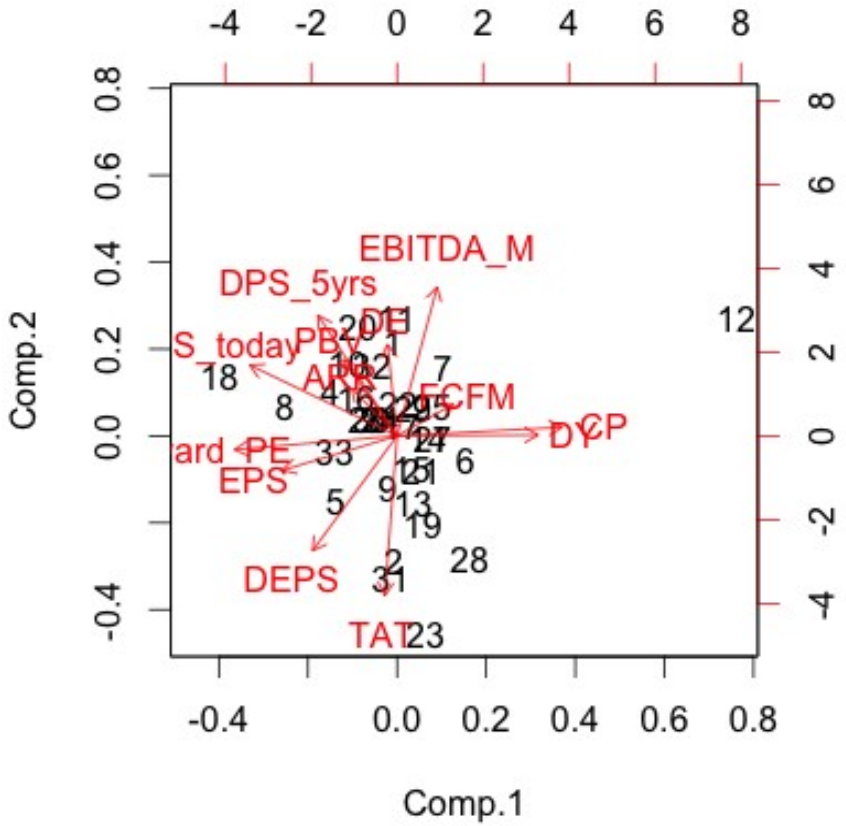
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.1895     1.0078   0.188  0.85231
yv1          -2.6271     0.5452  -4.819 5.42e-05 ***
yv2           4.5379     0.6246   7.265 1.03e-07 ***
yv3           0.2931     0.7350   0.399  0.69330
yv4           2.5331     0.8329   3.041  0.00533 **
yv5          -2.2889     0.9132  -2.506  0.01879 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.701 on 26 degrees of freedom
Multiple R-squared:  0.7791,    Adjusted R-squared:  0.7366
F-statistic: 18.34 on 5 and 26 DF,  p-value: 8.565e-08
```

Residual Plots w/out ADM



Biplot – Traded Companies



What's Next?

- Understand the principal components in terms of the data and from a financial perspective
- Examine what variables (or sectors) have positive or negative correlations with each principal component
- Run the data for more than one quarter to see whether our model does a good job in predicting percent change for future quarters